

Statistics & Probability

ANSWER KEY



Mean, median, mode, and range

- 1 Find the mean, median, and range of the data set: 12, 18, 15, 22, 9, 18.

Mean = $\frac{12 + 18 + 15 + 22 + 9 + 18}{6} = \frac{94}{6} \approx 15.67$. Sorted: 9, 12, 15, 18, 18, 22. Median (average of the two middle values): $\frac{15 + 18}{2} = 16.5$. Range: $22 - 9 = 13$.

- 2 A data set of 5 values has a mean of 14. If four of the values are 10, 12, 16, 19, find the fifth value.

Sum of all 5 values: $14 \times 5 = 70$. Sum of known values: $10 + 12 + 16 + 19 = 57$. Fifth value: $70 - 57 = 13$.

Weighted averages

- 3 A class of 20 students has an average test score of 78, and another class of 30 students has an average of 85. Find the combined average score of all 50 students.

Total points from class 1: $20 \times 78 = 1560$. Total points from class 2: $30 \times 85 = 2550$. Combined total: $1560 + 2550 = 4110$. Combined average: $\frac{4110}{50} = 82.2$.

Basic probability

- 4 A bag contains 6 red, 4 blue, and 5 green marbles. Find the probability of drawing a blue marble.

Total marbles: $6 + 4 + 5 = 15$. $P(\text{blue}) = \frac{4}{15}$.

- 5 A fair six-sided die is rolled twice. Find the probability of rolling a 6 both times.

Since the rolls are independent: $P(6 \text{ and } 6) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$.

Combinations and permutations

- 6 In how many ways can a committee of 3 people be chosen from a group of 8, using combinations?

$$\binom{8}{3} = \frac{8!}{3!5!} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56.$$

- 7 In how many ways can first, second, and third place be awarded among 6 runners, using permutations?

$$P(6, 3) = \frac{6!}{3!} = 6 \times 5 \times 4 = 120.$$

Conditional probability and two-way tables

- 8 A survey of 250 students found: 90 like math and science, 60 like only math, 50 like only science, 50 like neither. Find the probability a student likes science, given that they like math.

Total who like math: $90 + 60 = 150$. Of those, 90 also like science. $P(\text{science} \mid \text{math}) = \frac{90}{150} = 0.6$, or 60%.

Standard deviation and data spread

- 9 Two data sets have the same mean, but Data Set A has standard deviation 2 and Data Set B has standard deviation 9. Explain what this tells you about the two data sets.

A smaller standard deviation (Data Set A) means the values are clustered tightly around the mean, while a larger standard deviation (Data Set B) means the values are much more spread out — even though both sets share the same central tendency (mean).

Reading and interpreting data displays

- 10 A bar graph shows quarterly sales: Q1: 40, Q2: 55, Q3: 65, Q4: 50 (in thousands of dollars). Find the total annual sales and the percent of annual sales that occurred in Q3.

Total annual sales: $40 + 55 + 65 + 50 = 210$ thousand dollars. Percent from Q3: $\frac{65}{210} \times 100\% \approx 31\%$.

Expected value

- 11 A game costs 5 dollars to play. There is a 20% chance of winning 20 dollars and an 80% chance of winning nothing. Find the expected value of playing the game, and determine whether it is a fair game.

Expected winnings: $0.20(20) + 0.80(0) = 4$ dollars. Since the cost to play is 5 dollars but the expected winnings are only 4 dollars, the expected net value is $4 - 5 = -1$ dollar — the game is not fair; on average, a player loses 1 dollar per play.

Multi-part probability and statistics problem

- 12 This question has two parts. A jar contains 5 red and 3 blue chips. Two chips are drawn without replacement. (a) Find the probability that both chips are red. (b) Find the probability that the two chips are different colors.

(a) $P(\text{first red}) = \frac{5}{8}$. $P(\text{second red} \mid \text{first red}) = \frac{4}{7}$. $P(\text{both red}) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$. (b)
 $P(\text{red then blue}) = \frac{5}{8} \times \frac{3}{7} = \frac{15}{56}$. $P(\text{blue then red}) = \frac{3}{8} \times \frac{5}{7} = \frac{15}{56}$.
 $P(\text{different colors}) = \frac{15}{56} + \frac{15}{56} = \frac{30}{56} = \frac{15}{28}$.

The complement rule and 'at least' probability

- 13** A fair coin is flipped 3 times. Find the probability of getting at least one heads, using the complement rule.

$$P(\text{no heads}) = P(\text{all tails}) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}. P(\text{at least one heads}) = 1 - \frac{1}{8} = \frac{7}{8}.$$

Quartiles and the five-number summary

- 14** A data set has the five-number summary: minimum 5, Q1 = 12, median 18, Q3 = 25, maximum 40. Find the interquartile range (IQR), and state what it measures.

$IQR = Q3 - Q1 = 25 - 12 = 13$. The IQR measures the spread of the middle 50% of the data, and is often used to identify outliers, since it is less sensitive to extreme values than the full range.

Independent events and the multiplication rule

- 15** A spinner has 4 equal sections (red, blue, green, yellow), and a separate fair die is rolled. Find the probability of spinning red AND rolling a number greater than 4.

$$P(\text{red}) = \frac{1}{4}. P(\text{greater than 4}) = P(5 \text{ or } 6) = \frac{2}{6} = \frac{1}{3}. \text{ Since independent: } P(\text{red and } > 4) = \frac{1}{4} \times \frac{1}{3} = \frac{1}{12}.$$

A multi-part statistics problem with grouped data

- 16** This question has two parts. A teacher records test scores for two classes: Class A (25 students, mean 82) and Class B (15 students, mean 90). (a) Find the combined mean score for all 40 students. (b) If 5 more students join Class A with an average score of 70, find the new combined mean for all 45 students in the combined group.

(a) Total from A: $25(82) = 2050$. Total from B: $15(90) = 1350$. Combined total: $2050 + 1350 = 3400$. Combined mean: $\frac{3400}{40} = 85$. (b) New students' total: $5(70) = 350$. New grand total: $3400 + 350 = 3750$. New mean: $\frac{3750}{45} \approx 83.3$.

Probability using a tree diagram approach

- 17** A box has 4 white and 6 black balls. Two balls are drawn without replacement. Find the probability both balls are white.

$$P(\text{first white}) = \frac{4}{10}. P(\text{second white} \mid \text{first white}) = \frac{3}{9}. P(\text{both white}) = \frac{4}{10} \times \frac{3}{9} = \frac{12}{90} = \frac{2}{15}.$$

Multiple choice — easy

- 18** A bag contains 7 red marbles, 9 blue marbles, and 8 green marbles. Find the probability of drawing a red marble.

- a. $\frac{1}{4}$
- b. $\frac{3}{8}$
- c. $\frac{7}{24}$ ✓
- d. $\frac{1}{3}$

Total marbles: $7 + 9 + 8 = 24$. $P(\text{red}) = \frac{7}{24} = \frac{7}{24}$.

- 19** A bag contains 11 red marbles, 8 blue marbles, and 3 green marbles. Find the probability of drawing a green marble.

- a. $\frac{2}{11}$
- b. $\frac{4}{11}$
- c. $\frac{1}{2}$
- d. $\frac{3}{22}$ ✓

Total marbles: $11 + 8 + 3 = 22$. $P(\text{green}) = \frac{3}{22} = \frac{3}{22}$.

- 20** A bag contains 5 black marbles, 4 white marbles, and 10 yellow marbles. Find the probability of drawing a black marble.

- a. $\frac{10}{19}$
- b. $\frac{6}{19}$
- c. $\frac{5}{19}$ ✓
- d. $\frac{4}{19}$

Total marbles: $5 + 4 + 10 = 19$. $P(\text{black}) = \frac{5}{19} = \frac{5}{19}$.

- 21 A bag contains 10 red marbles, 7 blue marbles, and 6 green marbles. Find the probability of drawing a red marble.

a. $\frac{10}{23}$ ✓

b. $\frac{7}{23}$

c. $\frac{6}{23}$

d. $\frac{11}{23}$

Total marbles: $10 + 7 + 6 = 23$. $P(\text{red}) = \frac{10}{23} = \frac{10}{23}$.

- 22 A bag contains 2 orange marbles, 10 purple marbles, and 12 silver marbles. Find the probability of drawing a purple marble.

a. $\frac{1}{2}$

b. $\frac{1}{12}$

c. $\frac{5}{12}$ ✓

d. $\frac{11}{24}$

Total marbles: $2 + 10 + 12 = 24$. $P(\text{purple}) = \frac{10}{24} = \frac{5}{12}$.

- 23 A bag contains 6 red marbles, 11 blue marbles, and 8 green marbles. Find the probability of drawing a green marble.

a. $\frac{8}{25}$ ✓

b. $\frac{6}{25}$

c. $\frac{11}{25}$

d. $\frac{9}{25}$

Total marbles: $6 + 11 + 8 = 25$. $P(\text{green}) = \frac{8}{25} = \frac{8}{25}$.

- 24** A bag contains 4 orange marbles, 4 purple marbles, and 11 silver marbles. Find the probability of drawing an orange marble.

- a. $\frac{5}{19}$
- b. $\frac{3}{19}$
- c. $\frac{4}{19}$ ✓
- d. $\frac{11}{19}$

Total marbles: $4 + 4 + 11 = 19$. $P(\text{orange}) = \frac{4}{19} = \frac{4}{19}$.

- 25** A bag contains 5 red marbles, 7 blue marbles, and 6 green marbles. Find the probability of drawing a green marble.

- a. $\frac{7}{18}$
- b. $\frac{2}{9}$
- c. $\frac{5}{18}$
- d. $\frac{1}{3}$ ✓

Total marbles: $5 + 7 + 6 = 18$. $P(\text{green}) = \frac{6}{18} = \frac{1}{3}$.

- 26** A bag contains 4 black marbles, 6 white marbles, and 5 yellow marbles. Find the probability of drawing a white marble.

- a. $\frac{4}{15}$
- b. $\frac{1}{3}$
- c. $\frac{2}{5}$ ✓
- d. $\frac{7}{15}$

Total marbles: $4 + 6 + 5 = 15$. $P(\text{white}) = \frac{6}{15} = \frac{2}{5}$.

- 27 A bag contains 5 black marbles, 12 white marbles, and 9 yellow marbles. Find the probability of drawing a white marble.

- a. $\frac{1}{2}$
- b. $\frac{5}{26}$
- c. $\frac{6}{13}$ ✓
- d. $\frac{9}{26}$

Total marbles: $5 + 12 + 9 = 26$. $P(\text{white}) = \frac{12}{26} = \frac{6}{13}$.

- 28 A bag contains 9 black marbles, 10 white marbles, and 4 yellow marbles. Find the probability of drawing a yellow marble.

- a. $\frac{4}{23}$ ✓
- b. $\frac{10}{23}$
- c. $\frac{9}{23}$
- d. $\frac{5}{23}$

Total marbles: $9 + 10 + 4 = 23$. $P(\text{yellow}) = \frac{4}{23} = \frac{4}{23}$.

- 29 A bag contains 5 orange marbles, 9 purple marbles, and 7 silver marbles. Find the probability of drawing a silver marble.

- a. $\frac{8}{21}$
- b. $\frac{3}{7}$
- c. $\frac{5}{21}$
- d. $\frac{1}{3}$ ✓

Total marbles: $5 + 9 + 7 = 21$. $P(\text{silver}) = \frac{7}{21} = \frac{1}{3}$.

- 30 A bag contains 7 red marbles, 10 blue marbles, and 9 green marbles. Find the probability of drawing a blue marble.

- a. $\frac{7}{26}$
- b. $\frac{9}{26}$
- c. $\frac{11}{26}$
- d. $\frac{5}{13}$ ✓

Total marbles: $7 + 10 + 9 = 26$. $P(\text{blue}) = \frac{10}{26} = \frac{5}{13}$.

- 31 A bag contains 6 black marbles, 5 white marbles, and 6 yellow marbles. Find the probability of drawing a black marble.

- a. $\frac{5}{17}$
- b. $\frac{4}{17}$
- c. $\frac{7}{17}$
- d. $\frac{6}{17}$ ✓

Total marbles: $6 + 5 + 6 = 17$. $P(\text{black}) = \frac{6}{17} = \frac{6}{17}$.

- 32 A bag contains 11 orange marbles, 11 purple marbles, and 12 silver marbles. Find the probability of drawing a silver marble.

- a. $\frac{5}{17}$
- b. $\frac{13}{34}$
- c. $\frac{11}{34}$
- d. $\frac{6}{17}$ ✓

Total marbles: $11 + 11 + 12 = 34$. $P(\text{silver}) = \frac{12}{34} = \frac{6}{17}$.

Multiple choice — medium

33 In how many ways can 4 distinct positions be awarded among 6 people, using permutations?

a. 356

b. 15

c. 360 ✓

d. 384

$$P(6, 4) = \frac{6!}{(2)!} = 360.$$

34 In how many ways can a committee of 2 people be chosen from a group of 8, using combinations?

a. 30

b. 56

c. 26

d. 28 ✓

$$\binom{8}{2} = \frac{8!}{2! (6)!} = 28.$$

35 In how many ways can a committee of 2 people be chosen from a group of 10, using combinations?

a. 43

b. 45 ✓

c. 47

d. 90

$$\binom{10}{2} = \frac{10!}{2! (8)!} = 45.$$

36 In how many ways can 2 distinct positions be awarded among 7 people, using permutations?

a. 42 ✓

b. 44

c. 21

d. 40

$$P(7, 2) = \frac{7!}{(5)!} = 42.$$

37 In how many ways can 4 distinct positions be awarded among 9 people, using permutations?

a. 3048

b. 3024 ✓

c. 3020

d. 126

$$P(9, 4) = \frac{9!}{(9-4)!} = 3024.$$

38 In how many ways can a committee of 2 people be chosen from a group of 7, using combinations?

a. 21 ✓

b. 42

c. 23

d. 19

$$\binom{7}{2} = \frac{7!}{2!(7-2)!} = 21.$$

39 In how many ways can 4 distinct positions be awarded among 7 people, using permutations?

a. 35

b. 836

c. 864

d. 840 ✓

$$P(7, 4) = \frac{7!}{(7-4)!} = 840.$$

40 In how many ways can a committee of 3 people be chosen from a group of 9, using combinations?

a. 84 ✓

b. 90

c. 81

d. 504

$$\binom{9}{3} = \frac{9!}{3!(6)!} = 84.$$

41 In how many ways can 4 distinct positions be awarded among 11 people, using permutations?

a. 7916

b. 330

c. 7920 ✓

d. 7944

$$P(11, 4) = \frac{11!}{(7)!} = 7920.$$

42 In how many ways can 4 distinct positions be awarded among 5 people, using permutations?

a. 144

b. 116

c. 120 ✓

d. 5

$$P(5, 4) = \frac{5!}{(1)!} = 120.$$

43 In how many ways can 3 distinct positions be awarded among 11 people, using permutations?

a. 990 ✓

b. 987

c. 165

d. 996

$$P(11, 3) = \frac{11!}{(8)!} = 990.$$

44 In how many ways can 4 distinct positions be awarded among 10 people, using permutations?

a. 210

b. 5064

c. 5036

d. 5040 ✓

$$P(10, 4) = \frac{10!}{(6)!} = 5040.$$

45 In how many ways can a committee of 3 people be chosen from a group of 6, using combinations?

a. 17

b. 26

c. 120

d. 20 ✓

$$\binom{6}{3} = \frac{6!}{3!(3)!} = 20.$$

46 In how many ways can 2 distinct positions be awarded among 12 people, using permutations?

a. 66

b. 132 ✓

c. 130

d. 134

$$P(12, 2) = \frac{12!}{(10)!} = 132.$$

47 In how many ways can a committee of 4 people be chosen from a group of 5, using combinations?

- a. 1
- b. 29
- c. 5 ✓
- d. 120

$$\binom{5}{4} = \frac{5!}{4!(1)!} = 5.$$

Multiple choice — hard

48 A class of 34 students has an average test score of 76, and another class of 25 students has an average of 97. Find the combined average score of all 59 students.

- a. 84.90 ✓
- b. 86.50
- c. 86.90
- d. 82.90

Total from class 1: $34 \times 76 = 2584$. Total from class 2: $25 \times 97 = 2425$. Combined total: $2584 + 2425 = 5009$. Combined average: $5009/59 = 84.90$.

49 A class of 18 students has an average test score of 69, and another class of 37 students has an average of 93. Find the combined average score of all 55 students.

- a. 81.00
- b. 85.15 ✓
- c. 83.15
- d. 87.15

Total from class 1: $18 \times 69 = 1242$. Total from class 2: $37 \times 93 = 3441$. Combined total: $1242 + 3441 = 4683$. Combined average: $4683/55 = 85.15$.

- 50 A class of 32 students has an average test score of 73, and another class of 25 students has an average of 88. Find the combined average score of all 57 students.
- a. 80.50
 - b. 77.58
 - c. 81.58
 - d. 79.58 ✓

Total from class 1: $32 \times 73 = 2336$. Total from class 2: $25 \times 88 = 2200$. Combined total: $2336 + 2200 = 4536$. Combined average: $4536/57 = 79.58$.

- 51 A class of 34 students has an average test score of 87, and another class of 25 students has an average of 74. Find the combined average score of all 59 students.
- a. 80.50
 - b. 81.49 ✓
 - c. 79.49
 - d. 83.49

Total from class 1: $34 \times 87 = 2958$. Total from class 2: $25 \times 74 = 1850$. Combined total: $2958 + 1850 = 4808$. Combined average: $4808/59 = 81.49$.

- 52 A class of 17 students has an average test score of 64, and another class of 10 students has an average of 95. Find the combined average score of all 27 students.
- a. 79.50
 - b. 75.48 ✓
 - c. 73.48
 - d. 77.48

Total from class 1: $17 \times 64 = 1088$. Total from class 2: $10 \times 95 = 950$. Combined total: $1088 + 950 = 2038$. Combined average: $2038/27 = 75.48$.

- 53 A box has 5 red and 3 white balls. Two balls are drawn without replacement. Find the probability that the first is red and the second is white.

- a. $\frac{89}{280}$
- b. $\frac{5}{8}$
- c. $\frac{15}{28}$
- d. $\frac{15}{56}$ ✓

$P(\text{first}) = \frac{5}{8}$. After removing one ball, $P(\text{second} \mid \text{first}) = \frac{3}{7}$. $P(\text{both}) = \frac{5}{8} \times \frac{3}{7} = \frac{15}{56}$.

- 54 A box has 8 red and 8 white balls. Two balls are drawn without replacement. Find the probability that the first is white and the second is red.

- a. $\frac{1}{2}$
- b. $\frac{8}{15}$
- c. $\frac{19}{60}$
- d. $\frac{4}{15}$ ✓

$P(\text{first}) = \frac{1}{2}$. After removing one ball, $P(\text{second} \mid \text{first}) = \frac{8}{15}$. $P(\text{both}) = \frac{1}{2} \times \frac{8}{15} = \frac{4}{15}$.

- 55 A box has 7 red and 3 white balls. Two balls are drawn without replacement. Find the probability that the first is white and the second is red.

- a. $\frac{3}{10}$
- b. $\frac{7}{15}$
- c. $\frac{7}{30}$ ✓
- d. $\frac{17}{60}$

$P(\text{first}) = \frac{3}{10}$. After removing one ball, $P(\text{second} \mid \text{first}) = \frac{7}{9}$. $P(\text{both}) = \frac{3}{10} \times \frac{7}{9} = \frac{7}{30}$.

- 56 A box has 8 red and 3 white balls. Two balls are drawn without replacement. Find the probability that the first is white and the second is red.

- a. $\frac{59}{220}$
- b. $\frac{3}{11}$
- c. $\frac{24}{55}$
- d. $\frac{12}{55}$ ✓

$P(\text{first}) = \frac{3}{11}$. After removing one ball, $P(\text{second} \mid \text{first}) = \frac{4}{5}$. $P(\text{both}) = \frac{3}{11} \times \frac{4}{5} = \frac{12}{55}$.

- 57 A box has 6 red and 8 white balls. Two balls are drawn without replacement. Find the probability that both are red.

- a. $\frac{3}{7}$
- b. $\frac{15}{91}$ ✓
- c. $\frac{30}{91}$
- d. $\frac{391}{1820}$

$P(\text{first}) = \frac{3}{7}$. After removing one ball, $P(\text{second} \mid \text{first}) = \frac{5}{13}$. $P(\text{both}) = \frac{3}{7} \times \frac{5}{13} = \frac{15}{91}$.

- 58 A fair coin is flipped 3 times. Find the probability of getting at least one heads, using the complement rule.

- a. $\frac{39}{40}$
- b. $\frac{1}{8}$
- c. $\frac{7}{8}$ ✓
- d. $\frac{1}{3}$

$P(\text{all tails}) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$. By the complement rule: $1 - \frac{1}{8} = \frac{7}{8}$.

- 59 A fair six-sided die is rolled 4 times. Find the probability of rolling at least one 6, using the complement rule.

a. $\frac{4003}{6480}$

b. $\frac{671}{1296}$ ✓

c. $\frac{625}{1296}$

d. $\frac{1}{4}$

$$P(\text{no 6s}) = \left(\frac{5}{6}\right)^4 = \frac{625}{1296}. \text{ By the complement rule: } 1 - \frac{625}{1296} = \frac{671}{1296}.$$

- 60 A fair coin is flipped 5 times. Find the probability of getting at least one heads, using the complement rule.

a. $\frac{1}{32}$

b. $\frac{139}{160}$

c. $\frac{31}{32}$ ✓

d. $\frac{1}{5}$

$$P(\text{all tails}) = \left(\frac{1}{2}\right)^5 = \frac{1}{32}. \text{ By the complement rule: } 1 - \frac{1}{32} = \frac{31}{32}.$$

- 61 A fair six-sided die is rolled 6 times. Find the probability of rolling at least one 6, using the complement rule.

a. $\frac{15625}{46656}$

b. $\frac{31031}{46656}$ ✓

c. $\frac{178483}{233280}$

d. $\frac{1}{6}$

$$P(\text{no 6s}) = \left(\frac{5}{6}\right)^6 = \frac{15625}{46656}. \text{ By the complement rule: } 1 - \frac{15625}{46656} = \frac{31031}{46656}.$$

62 A fair six-sided die is rolled 2 times. Find the probability of rolling at least one 6, using the complement rule.

a. $\frac{1}{2}$

b. $\frac{25}{36}$

c. $\frac{11}{36}$ ✓

d. $\frac{73}{180}$

$P(\text{no 6s}) = \left(\frac{5}{6}\right)^2 = \frac{25}{36}$. By the complement rule: $1 - \frac{25}{36} = \frac{11}{36}$.